

Refutation of Conjecture 00000001367: The Borel Chromatic Number of the $\mathbb{Z}^2$ Grid is 2, Not 5

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Abstract

We refute conjecture 00000001367, which asserts that the Borel chromatic number of the Cayley graph of $\mathbb{Z}^2$ with generators $\{\pm e_1, \pm e_2\}$ is exactly 5 while its classical chromatic number is 4, the Borel constraint allegedly forcing one extra color. The graph in question is the square grid, which is bipartite; hence its classical chromatic number is 2, not 4. Since $\mathbb{Z}^2$ is countable and carries the discrete σ -algebra, every subset of the vertex set is Borel, so the parity 2-coloring is Borel; hence the Borel chromatic number is also 2, not 5. Both asserted values are wrong, and the Borel constraint forces no extra color here.

1 The conjecture

The conjecture under consideration reads verbatim:

Definition: The Borel chromatic number of a Borel graph on a standard Borel space is the least number of colors in a Borel-measurable proper coloring. Conjecture: The Borel chromatic number of the Cayley graph of $\mathbb{Z}^2$ (generators $\pm e_1, \pm e_2$) is exactly 5; the classical chromatic number is 4, but the Borel constraint forces one extra color.

We show that both numerical assertions (4 and 5) are false.

2 The graph is the square grid

Definition 1 (Cayley graph). Let G be a group and $S \subseteq G$ a symmetric generating set (i.e. $S = S^{-1}$ and $e \notin S$). The *Cayley graph* $\text{Cay}(G, S)$ has vertex set G and an (undirected) edge between g and h iff $h = gs$ for some $s \in S$.

Take $G = \mathbb{Z}^2$ (written additively) and $S = \{\pm e_1, \pm e_2\}$, where $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Since G is abelian and $S = -S$, the edge relation becomes: $p, q \in \mathbb{Z}^2$ are adjacent iff

$$q = p + e_1, \quad q = p - e_1, \quad q = p + e_2, \quad \text{or} \quad q = p - e_2.$$

Equivalently, writing $p = (i, j)$ and $q = (i', j')$, two lattice points are adjacent iff $|i - i'| + |j - j'| = 1$. This is precisely the *square grid*.

3 Bipartiteness and the classical chromatic number

Definition 2 (Proper coloring, chromatic number). A *proper coloring* of a graph $G = (V, E)$ is a map $c: V \rightarrow C$ such that $c(p) \neq c(q)$ whenever $\{p, q\} \in E$. The *chromatic number* $\chi(G)$ is the least cardinality $|C|$ for which a proper coloring exists.

Theorem 3 (Bipartiteness). *The square grid $G = \text{Cay}(\mathbb{Z}^2, \{\pm e_1, \pm e_2\})$ is bipartite. Explicitly, the map*

$$c_0: \mathbb{Z}^2 \rightarrow \{0, 1\}, \quad c_0(i, j) := (i + j) \bmod 2,$$

is a proper 2-coloring.

Proof. Let $p = (i, j)$ and let q be adjacent to p . There are four cases.

- $q = (i + 1, j)$: then $q_1 + q_2 = i + j + 1$, so $c_0(q) \equiv (i + j + 1) \equiv c_0(p) + 1 \pmod{2}$.
- $q = (i - 1, j)$: then $q_1 + q_2 = i + j - 1 \equiv c_0(p) + 1 \pmod{2}$.
- $q = (i, j + 1)$: then $q_1 + q_2 = i + j + 1 \equiv c_0(p) + 1 \pmod{2}$.
- $q = (i, j - 1)$: then $q_1 + q_2 = i + j - 1 \equiv c_0(p) + 1 \pmod{2}$.

In every case $c_0(q) \equiv c_0(p) + 1 \pmod{2}$, hence $c_0(q) \neq c_0(p)$. Thus c_0 is proper, and the bipartition is given by the two parity classes of $i + j$. $\square$

Corollary 4 (Classical chromatic number). $\chi(G) = 2$. *In particular $\chi(G) \neq 4$.*

Proof. By Theorem 3, c_0 is a proper 2-coloring, so $\chi(G) \leq 2$. On the other hand G has at least one edge, for instance $(0, 0) - (1, 0)$; no graph with an edge admits a proper 1-coloring, so $\chi(G) \geq 2$. Hence $\chi(G) = 2$. $\square$

4 Countability and the Borel chromatic number

We recall the relevant definitions.

Definition 5 (Borel graph and Borel chromatic number). A *Borel graph* on a standard Borel space $(X, \mathcal{B})$ is a pair (X, E) where $E \subseteq X \times X$ is a symmetric, irreflexive Borel set. Its *Borel chromatic number* $\chi_B(G)$ is the least cardinality of a standard Borel space C admitting a Borel-measurable proper coloring $c: X \rightarrow C$.

The vertex set $\mathbb{Z}^2$ is countable. We equip it with the discrete σ -algebra $\mathcal{B} = \mathcal{P}(\mathbb{Z}^2)$, i.e. every subset is measurable. The pair $(\mathbb{Z}^2, \mathcal{P}(\mathbb{Z}^2))$ is a standard Borel space: it is Borel-isomorphic to a countable discrete Polish space (for example, transport a bijection $\mathbb{Z}^2 \rightarrow \mathbb{N}$). Because the space is discrete, the edge set is all of $E \subseteq V \times V$, a countable union of singletons, hence Borel, and G is a Borel graph. The following lemma is the key point.

Lemma 6 (Every subset is Borel). *On the countable discrete space $\mathbb{Z}^2$, every subset is Borel. More generally, every map from $\mathbb{Z}^2$ to any standard Borel space is Borel measurable.*

Proof. For each $p \in \mathbb{Z}^2$ the singleton $\{p\}$ is measurable (discrete σ -algebra), and a countable union of measurable sets is measurable, so every $A = \bigcup_{p \in A} \{p\}$ is Borel. For measurability of a map $f: \mathbb{Z}^2 \rightarrow Y$ with Y standard Borel, it suffices that $f^{-1}(B)$ be Borel for every Borel $B \subseteq Y$; since every subset of $\mathbb{Z}^2$ is Borel, this holds trivially. $\square$

Theorem 7 (Borel chromatic number). $\chi_B(G) = 2$. In particular $\chi_B(G) \neq 5$.

Proof. Upper bound. By Lemma 6 the parity coloring $c_0: \mathbb{Z}^2 \rightarrow \{0, 1\}$ of Theorem 3 is Borel measurable (its codomain $\{0, 1\}$ is a standard Borel space), and it is proper. Hence $\chi_B(G) \leq 2$.

Lower bound. As in Corollary 4, G has an edge, and any proper coloring must assign distinct colors to the two endpoints of an edge, so no 1-coloring is proper. Therefore $\chi_B(G) \geq 2$.

Combining, $\chi_B(G) = 2$. □

Remark 8 (No extra color is forced). Since $\chi(G) = \chi_B(G) = 2$, the Borel constraint does not force an additional color in this example. The phenomenon the conjecture invokes is real in other settings—for instance, Borel graphs of bounded degree can have finite Borel chromatic number strictly larger than their classical one—but it does not occur for the square grid, whose bipartition is already Borel because the space is countable and discrete.

5 Conclusion

Both values asserted by conjecture 00000001367 are false:

$$\chi(G) = 2 \quad (\text{not } 4), \quad \chi_B(G) = 2 \quad (\text{not } 5).$$

The Cayley graph of $\mathbb{Z}^2$ with generators $\{\pm e_1, \pm e_2\}$ is the square grid, which is bipartite via the parity of $i + j$; and because $\mathbb{Z}^2$ is countable and discrete, every subset—in particular the parity classes—is Borel, so the very same 2-coloring witnesses $\chi_B \leq 2$. Hence the conjecture is false in both of its numerical assertions, and its explanation that “the Borel constraint forces one extra color” does not apply here. The combinatorial part of this refutation is mechanically verified in core Lean 4 (see `lean4/`); the Borel measurability step is the standard argument of Lemma 6 and is not formalised in Lean.